

Design and Implementation of an Automated Wind-Tunnel Platform for Heatsink Thermal Characterisation

Alexander Groenvynck

Academic year 2025–2026

Bachelor in Energy Technology — Odisee University of Applied Sciences, Ghent

Exchange: University of Girona Supervisor: Lino Montoro Moreno

Table of Contents

1. The Problem
2. Research Questions
3. System Overview
4. Wind Tunnel & Hardware
5. Control Strategy
6. Software Architecture
7. The Automated Workflow
8. Results & Validation
9. Safety
10. Conclusion & Future Work

The Problem

Manual heatsink testing is:

- Slow
- Error-prone
- Unsafe — live heater

Goal: one button → results CSV.

[PHOTO]

Complete assembled system
on the lab bench

Research Questions

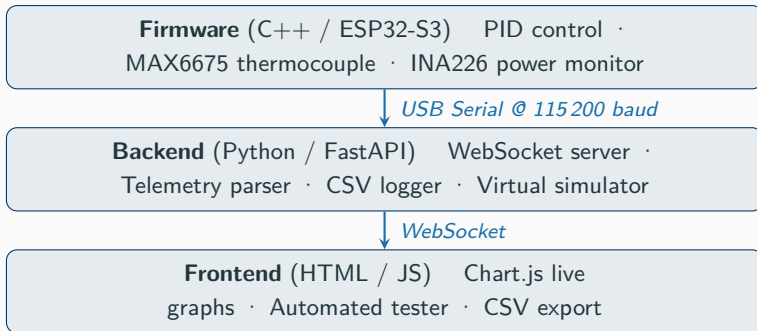
RQ1 Hold set-point temperatures with enough *stability*?

RQ2 Detect thermal equilibrium *automatically*?

RQ3 Reduce the workflow to a *single action* + guarantee safe shutdown?

RQ4 Produce a metric that *distinguishes heatsinks*?

System Overview — Three-Tier Architecture



Wind Tunnel Construction

- Chipboard enclosure, fan at one end
- **Hexagonal honeycomb grid** after the fan
 - Breaks up fan swirl
 - Uniform, laminar-like flow
 - → fair, repeatable comparison
- Heater module at test-section centre

[PHOTO]

Wind tunnel exterior
— full view

[PHOTO]

Fan end with HEX grid
close-up

The Heater Module

Stack (bottom → top):

1. Insulation
2. Steel backing plate
3. **Ceramic PTC heater**
4. **Copper spreader plate**
5. Heatsink — wire clip

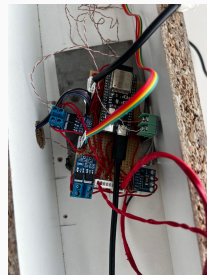
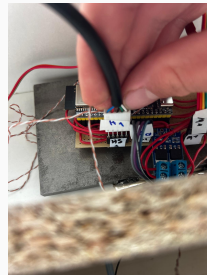
Thermocouple in drilled hole in copper plate.



Electronics & Sensors

- **ESP32-S3** — runs firmware
- **MAX6675** — thermocouple (SPI)
- **INA226** — power monitor (I²C)
- **2× MOSFETs** — heater + fan
- **Eventek** — 12 V PSU

All on a **perfboard** in the tunnel base.



Control Strategy: SMART Mode

Modes: AUTO · MANUAL · SMART

SMART (automated tests):

1. PID drives toward set-point
2. EMA filter on temperature
3. Stable $\pm 0.5^{\circ}\text{C} / 8\text{s} \rightarrow$ **HOLD** (lock PWM)
4. Backend confirms EqPWM convergence

PID $\rightarrow \pm 0.5^{\circ}\text{C} / 8\text{s}$ HOLD

Locked PWM = known steady-state power $P \rightarrow$ feeds $R_{th} = \Delta T / P$.

Thermal Metric: Thermal Resistance R_{th}

$$R_{th} = \frac{\Delta T}{P} \quad \left[\frac{^{\circ}\text{C}}{\text{W}} \right]$$

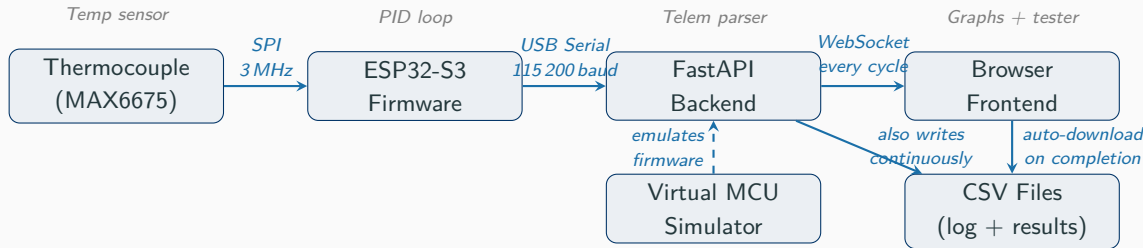
$\Delta T = T - T_{amb}$, P = heater power [W]

- Lower R_{th} = better heatsink
- Should be constant across set-points
- Same airflow $\rightarrow$ fair comparison

Intuition:

More **fins** = more convective area. Under forced airflow this beats raw conductivity — finned aluminium > solid copper.

Software Architecture: Data Flow



FastAPI / asyncio:

- `_serial_reader()` — read serial
- `_handle_line()` — parse `TELEM_RE`
- `broadcast()` — push WebSocket
- `sendCmd()` — SET / GET
- CSV logger (versioned schema)

Virtual MCU simulator

VIRTUAL port emulates firmware.

Built & tested the whole workflow before hardware existed.

Frontend: index.html

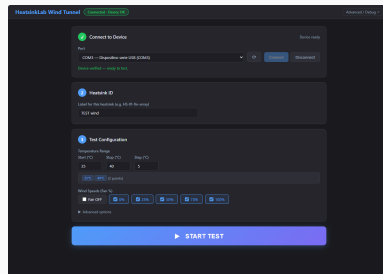
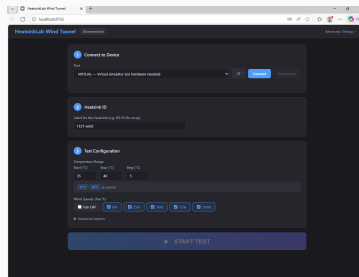
Single file — no build step

WebSocket handler:

- Updates Chart.js + readouts

Tester state machine (JS):

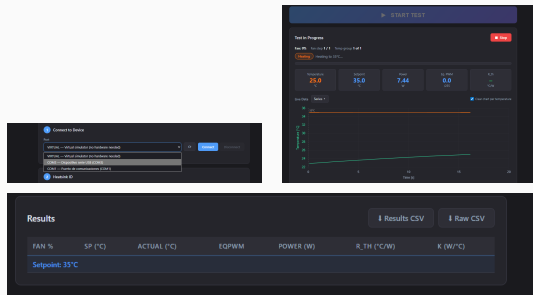
- Steps through set-points
- Waits for `eq_pwm` convergence
- Records row $\rightarrow$ next set-point
- Auto-downloads CSV at end



The Automated Workflow

1. Plug in ESP32
2. Open *HEATSINK TESTER*
3. Select port → Connect
4. Type heatsink ID
5. Press **Start Test**
6. **Wait** — runs automatically
7. CSV downloads on completion
8. Swap heatsink → repeat

No PID knowledge needed.



Results: R_{th} Comparison

Heatsink		SP (°C)	P (W)	R_{th} (°C/W)
HS-01 fin-array		40	7.6	2.38
HS-01 fin-array		50	11.6	2.42
HS-01 fin-array		60	15.9	2.38
HS-02	copper block	40	4.4	4.05
HS-02	copper block	50	7.0	3.99
HS-02	copper block	60	9.5	3.97

Representative values — illustrate output format.

Observations:

- R_{th} consistent across set-points
- Finned aluminium > solid copper
- Platform clearly separates them:
2.4 vs 4.0 °C/W

SET RUN OFF on every exit path:

1. Normal completion
2. Manual Stop button
3. Browser close / disconnect

All verified: INA226 power $\rightarrow$ 0.

Why it matters:

Unattended heater = fire / burn risk.

Automatic shutdown removes reliance on the student.

+ 60s fan cool-down before stop.

Limitations:

- Airspeed commanded, not measured
- No hardware safety watchdog
- Single thermocouple
- No formal repeatability study

Next steps:

1. Wire anemometer + SDP510
2. Hardware watchdog MCU
3. Repeatability study
4. Log ambient T + humidity
5. One-click installer

Conclusion

All four research questions answered:

- RQ1** PID + EMA: stable within $\pm 0.5^{\circ}\text{C}$
- RQ2** Dual-check equilibrium detection: reliable
- RQ3** One button; shutdown on all exit paths
- RQ4** Platform distinguishes heatsinks clearly

Contributions:

- Safe, usable lab automation
- **Software-first**: firmware locked
- Usable *without knowing PID*

[PHOTO]

Complete assembled system
on the lab bench

Thank you.

Questions?